PHYS1408 Exam 2 Notecard

Every 3×5 notecard equation, one card per problem type

$g = 9.8$ m/s². Each card's links go to the fully worked problems in the Exam 2 Worked Problems. For a minimum, round **up** to an answer choice that works; for a maximum, round **down**.

Curves

Worked examples: 6.1 · 6.2 · 6.3 · 6.4

$$v_{max} = \sqrt{gr\,\frac{\tan\theta + \mu_s}{1 - \mu_s \tan\theta}}$$

$$v_{min} = \sqrt{gr\,\frac{\tan\theta - \mu_s}{1 + \mu_s \tan\theta}}$$

- $g = 9.8$ m/s², r = radius of the curve (m).
- θ = bank angle from the horizontal; μ_s = coefficient of static friction (tires on road).
- **Flat curve:** $\theta = 0$ gives $v_{max} = \sqrt{\mu_s g r}$.
- **No friction:** $\mu_s = 0$ gives the single speed $v = \sqrt{gr\tan\theta}$.
- If $\tan\theta < \mu_s$, there is no v_{min} (the car can sit still on the bank).
- **Answer choices:** round a v_{max} **down** and a v_{min} **up** to a choice that works.

Free fall and roller coasters

Worked examples: 7.1 · 7.3

$$v_f = \sqrt{v_i^2 + 2g(h_i - h_f)}$$

- v_i, v_f = speed at the start / end (m/s); h_i, h_f = height at the start / end (m), from any reference level.
- Frictionless only. Starting from rest: $v_f = \sqrt{2g\,\Delta h}$, where Δh = the drop.
- If the object **rises** ($h_f > h_i$), the bracket is negative and it slows down.

Atwood's machine and hanging mass on a pulley

Worked examples: 7.2 · 10.4

$$v_f = \sqrt{v_i^2 + \frac{2gh(m_1 - m_2)}{m_1 + m_2 + \dfrac{I}{R^2}}}$$

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2 + \dfrac{I}{R^2}}$$

- m_1 = the hanging mass that **falls**; m_2 = the hanging mass that **rises**.
- h = distance each mass moves (m); v_i, v_f = their shared speed at the start / end.
- I, R = the pulley's moment of inertia and radius. Copy I from the problem; R cancels (disk $I/R^2 = \frac{1}{2}M$, ring $= M$, massless $= 0$).
- **One hanging mass** (string wrapped on a pulley): set $m_2 = 0$.
- String tension on the falling mass: $T = m_1(g - a)$.

Halfpipe

Worked examples: 7.4

$$h_f = h_i + \frac{v_i^2}{2g} - n\,\mu L$$

- h_i = starting height (m); h_f = height reached on the far side (m); v_i = starting speed (0 if starting from rest).
- μ = friction on the rough patch; L = length of the rough patch (m); n = number of times it crosses the patch.
- Frictionless: $h_f = h_i$. One crossing from rest: $h_f = h_i - \mu L$.
- **Number of full crossings before it stops** = the whole-number part of $\dfrac{h_i}{\mu L}$ (from rest).

Single block on an incline

Worked examples: 7.5 · 7.6 · 7.7 · 7.8 · R.2

Up the slope: $$v_f = \sqrt{v_i^2 - 2g\,\Delta x(\sin\theta + \mu_k \cos\theta)}$$

Down the slope: $$v_f = \sqrt{v_i^2 + 2g\,\Delta x(\sin\theta - \mu_k \cos\theta)}$$

Stops after (up): $$\Delta x = \frac{v_i^2}{2g(\sin\theta + \mu_k \cos\theta)}$$

Stops after (down): $$\Delta x = \frac{v_i^2}{2g(\mu_k \cos\theta - \sin\theta)}$$

Time: $$t = \frac{2\,\Delta x}{v_i + v_f}$$

- v_i, v_f = speed at the start / end (m/s); Δx = distance **along the slope** (m).
- θ = incline angle from the horizontal; μ_k = kinetic friction ($\mu_k = 0$ for frictionless).
- **Negative under the root** = it stops first. Use the stopping-distance line.
- Sliding down only stops if $\mu_k \cos\theta > \sin\theta$.
- Time with forces instead: $a = g(\sin\theta \pm \mu_k \cos\theta)$ (+ going up, − going down), then $t = \dfrac{|v_f - v_i|}{a}$.

Springs

Worked examples: 7.9 · 7.10

$$\tfrac{1}{2}kx^2 = \tfrac{1}{2}mv^2 + mgh$$

$$\text{Compression to stop a mass:} \quad x = \sqrt{\frac{2mgh + mv_i^2}{k}}$$

$$\text{Launch straight up:} \quad h = \frac{kx^2}{2mg}$$

$$\text{Launch along a level floor:} \quad v = x\sqrt{\frac{k}{m}}$$

- k = spring constant (N/m); x = how far the spring is compressed or stretched (m).
- m = the mass (kg); v = its speed when the spring is relaxed (m/s); h = height above (or drop to) the spring's position (m).
- For a launch, measure h from the **release point**. Frictionless only.

Power

Worked examples: 7.11

$$P = \frac{W}{t} = \frac{mgh}{t} = Fv$$

- P = power (W = J/s); W = work or energy (J); t = time (s).
- m = mass lifted (kg); h = height gained (m).
- F = force applied (N); v = speed it moves at (m/s), for a lift or pull at steady speed.

Incline + hanging mass + pulley

Worked examples: 7.12 · 7.13 · 10.6

$$v_f = \sqrt{v_i^2 + \frac{2g\,\Delta x\,B}{m_1 + m_2 + \dfrac{I}{R^2}}}$$

$$\text{Stops after:} \quad \Delta x = \frac{-\left(m_1 + m_2 + \dfrac{I}{R^2}\right) v_i^2}{2gB}$$

- m_1 = block on the incline; m_2 = hanging block ($m_2 = 0$ if nothing hangs); θ = incline angle; μ_k = friction between m_1 and the incline.
- Δx = distance each block moves (m); v_i, v_f = their shared speed at the start / end.
- I, R = pulley's moment of inertia and radius ($I = 0$ for a massless pulley; disk $I/R^2 = \frac{1}{2}M$).
- B = the "net pull" (kg), picked by direction of motion:
  - m_2 falls, m_1 slides **up**: $B = m_2 - m_1 \sin\theta - \mu_k m_1 \cos\theta$.
  - m_1 slides **down**, m_2 rises: $B = m_1 \sin\theta - m_2 - \mu_k m_1 \cos\theta$.
- $B > 0$: speeds up. $B < 0$: slows down; a negative number under the root means it stops first, so use the second line.

Loop-the-loop

Worked examples: 7.14

$$v_{top,\,min} = \sqrt{gR}$$

$$v_{bottom,\,min} = \sqrt{5gR}$$

$$h_{min} = \tfrac{5}{2}R$$

$$v_{top}^2 = v_{bottom}^2 - 4gR = 2g(h - 2R)$$

$$\text{Top:} \quad N = \frac{mv_{top}^2}{R} - mg$$

$$\text{Bottom:} \quad N = \frac{mv_{bottom}^2}{R} + mg$$

- R = loop radius (m); h = starting height above the bottom of the loop, starting from rest (m); m = mass (kg).
- v_{top}, v_{bottom} = speed at the top / bottom of the loop; N = normal force from the track (N).
- "Barely" makes it = $N = 0$ at the top. Launched from $h = \frac{5}{2}R$: $N_{bottom} = 6mg$.
- **Answer choices:** these are minimums, so round **up** to a choice that works.

Force from a momentum change

Worked examples: 9.1

$$F = \frac{m(v_f - v_i)}{t}$$

- m = mass (kg); v_i, v_f = velocity before / after **with signs** (reversing direction flips the sign); t = contact time (s).
- Bounces straight back at the same speed: $|F| = \dfrac{2mv}{t}$. Comes to a stop: $|F| = \dfrac{mv}{t}$.
- mv alone is momentum (kg · m/s), and $m(v_f - v_i)$ is the impulse (N · s); neither is a force.

Sticking collisions

Worked examples: 9.2

$$v_f = \frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2}$$

$$\frac{K_f}{K_i} = \frac{(m_1 + m_2)\, v_f^2}{m_1 v_{1i}^2 + m_2 v_{2i}^2}$$

- m_1, m_2 = the two masses; v_{1i}, v_{2i} = velocities before, **with signs** (pick a + direction).
- v_f = shared velocity after they stick; K_i, K_f = total kinetic energy before / after.
- One object at rest ($v_{2i} = 0$): $\dfrac{K_f}{K_i} = \dfrac{m_1}{m_1 + m_2}$.

Elastic collisions and the cue ball

Worked examples: 9.3 · 9.4

$$v_{1f} = \frac{(m_1 - m_2)v_{1i} + 2m_2 v_{2i}}{m_1 + m_2}$$

$$v_{2f} = v_{1i} + v_{1f} - v_{2i}$$

- m_1, m_2 = the two masses; v_{1i}, v_{2i} = velocities before; v_{1f}, v_{2f} = velocities after.
- **Every velocity has a sign.** Pick + (say, right); anything moving left is negative. A negative answer means it moves left.
- Either object can be "object 1".
- Equal masses, object 2 at rest (cue ball and 8-ball): $v_{1f} = 0$ and $v_{2f} = v_{1i}$.

Ballistic pendulum

Worked examples: 9.5

$$v = \frac{M + m}{m}\sqrt{2gh}$$

$$h = \frac{1}{2g}\left(\frac{mv}{M + m}\right)^2$$

$$\frac{K_f}{K_i} = \frac{m}{M + m}$$

- m = bullet mass; M = block mass; v = bullet speed before the hit (m/s); h = height the block swings up (m).
- Block + bullet speed right after the hit: $V = \dfrac{mv}{M + m} = \sqrt{2gh}$.
- K_f/K_i = fraction of the bullet's KE left in the swinging block.

Rotational kinematics

Worked examples: 10.1

$$\omega_f = \omega_i + \alpha t$$

$$\Delta\theta = \frac{\omega_i + \omega_f}{2}\, t$$

$$\omega_f^2 = \omega_i^2 + 2\alpha\, \Delta\theta$$

$$\Delta\theta = \omega_i t + \tfrac{1}{2}\alpha t^2$$

- ω_i, ω_f = angular velocity at the start / end (rad/s); α = constant angular acceleration (rad/s²).
- $\Delta\theta$ = angle turned (**radians**; 1 rev = 2π rad); t = time (s).
- Pick the equation that leaves out the one quantity you're not given and not asked for.

Parallel axis theorem

Worked examples: 10.2

$$I = I_{cm} + md^2$$

d = **distance from the object's CENTER to the new axis**, not from its end or edge.

Rod, axis a distance a from one end:

$$I = m\left[\frac{\ell^2}{12} + \left(\frac{\ell}{2} - a\right)^2\right]$$

Rod about one end ($a = 0$):

$$I = \tfrac{1}{3}m\ell^2$$

Disk about a point on its rim:

$$I = \tfrac{3}{2}MR^2$$

Ring about a point on its rim:

$$I = 2MR^2$$

Point mass at distance r:

$$I = mr^2$$

System of pieces (all about the same axis):

$$I_{total} = I_1 + I_2 + \cdots$$

- I = moment of inertia about the new axis (kg · m²); I_{cm} = about the object's own center (his I_0); m or M = the object's mass (kg).
- ℓ = rod length (m); a = how far the axis is from the **end** of the rod (m); R = radius (m); r = a point mass's distance from the axis (m).
- I_{cm} values: rod $\frac{1}{12}m\ell^2$, disk $\frac{1}{2}MR^2$, ring MR^2, sphere $\frac{2}{5}MR^2$, point mass 0.
- **Never** add md^2 to $\frac{1}{3}m\ell^2$. That value is already about the end.

Rolling acceleration

Worked examples: 10.3

$$a = \frac{g\sin\theta}{1 + \dfrac{I}{mR^2}}$$

- a = acceleration of the center down the incline (m/s²); θ = incline angle.
- I = the rolling object's moment of inertia; m, R = its mass and radius.
- Sphere $\frac{5}{7}g\sin\theta$; disk $\frac{2}{3}g\sin\theta$; ring $\frac{1}{2}g\sin\theta$; sliding $g\sin\theta$.

Rolling speed at the bottom

Worked examples: 10.5

$$v = \sqrt{\frac{2gh}{1 + \dfrac{I}{mR^2}}}$$

$$\omega = \frac{v}{R}$$

- v = speed of the center at the bottom (m/s); ω = spin rate at the bottom (rad/s); h = vertical drop (m), starting from rest.
- I = the object's moment of inertia; m, R = its mass and radius (R matters for ω, not for v).
- Ring $\sqrt{gh}$; disk $\sqrt{\frac{4}{3}gh}$; sphere $\sqrt{\frac{10}{7}gh}$; sliding $\sqrt{2gh}$.

Angular momentum

Worked examples: 11.1 · 11.2 · 11.3 · 11.4 · 11.5 · 11.6

$$\omega_f = \frac{I_1\omega_{1i} + I_2\omega_{2i} \pm mvr}{I_1 + I_2 + I_{added}}$$

- I_1, ω_{1i} = the spinning platform's moment of inertia and starting spin (merry-go-round disk: $I_1 = \frac{1}{2}MR^2$).
- I_2, ω_{2i} = a dropped object's moment of inertia and starting spin ($\omega_{2i} = 0$ if not spinning, **negative** if counter-rotating). Leave out if there isn't one.
- mvr = a person running and jumping on: m = their mass, v = their speed, r = distance from the axis where they land. + if running the same way as the rim, − if against it. Leave out if they just step on.
- I_{added} = what joins the platform: person or point mass mr^2, rod lying center-to-edge $\frac{1}{3}m\ell^2$, dropped disk $\frac{1}{2}mr^2$, ring mr^2.
- **Something moves in or out** (walking, pulling in arms): $\omega_f = \dfrac{I_i\,\omega_i}{I_f}$. Rim to center: $\omega_f = \dfrac{M + 2m}{M}\omega_i$.
- KE check: $K = \frac{1}{2}I\omega^2$. It drops in a "rotational collision" and rises when someone walks inward.

Projectile from a ledge

Worked examples: R.1

$$H = h + \frac{(v_0 \sin\theta)^2}{2g}$$

$$t_{top} = \frac{v_0 \sin\theta}{g}$$

- H = highest point above the ground (m); h = launch height above the ground (m; 0 if launched from the ground).
- v_0 = launch speed (m/s); θ = launch angle above the horizontal; t_{top} = time to reach the top (s).